early innovation in mechanical automation laid the groundwork for the sophisticated unmanned systems that define modern warfare.

Figure 2.4 illustrates the *Kettering Bug (1918)*, one of the earliest examples of an autonomous aerial torpedo, developed by Charles F. Kettering in the United States during World War I. The aircraft featured a lightweight wooden and fabric fuselage, twin wings, and a small piston engine that powered a wooden propeller. Its flight path was controlled by a mechanically pre-set guidance system, which used an internal gyroscope and a revolution counter to determine the distance flown before cutting the engine, allowing the drone to dive onto its target. Designed to deliver explosives without risking human pilots, the Kettering Bug marked a groundbreaking innovation in early unmanned flight. Though it never saw combat use, this design introduced the principles of automated navigation and pre-programed control, making it a direct precursor to modern cruise missile technology (Stoer, 2001).

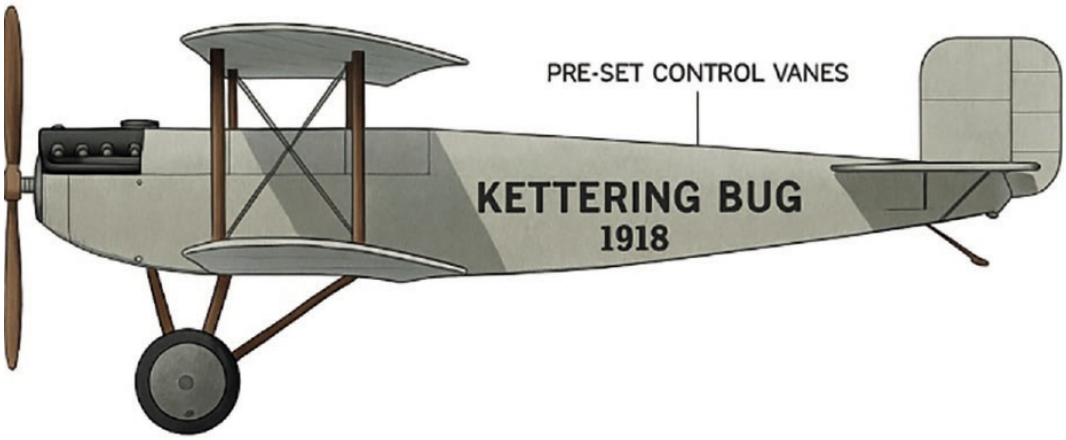


Figure 2.4 The Kettering Bug (1918) – a mechanically programed unmanned aerial torpedo.⏎

2.6.2.3 Queen Bee (1935)

The Queen Bee (1935), developed by the De Havilland Aircraft Company in the United Kingdom, stands as a landmark in the history of unmanned